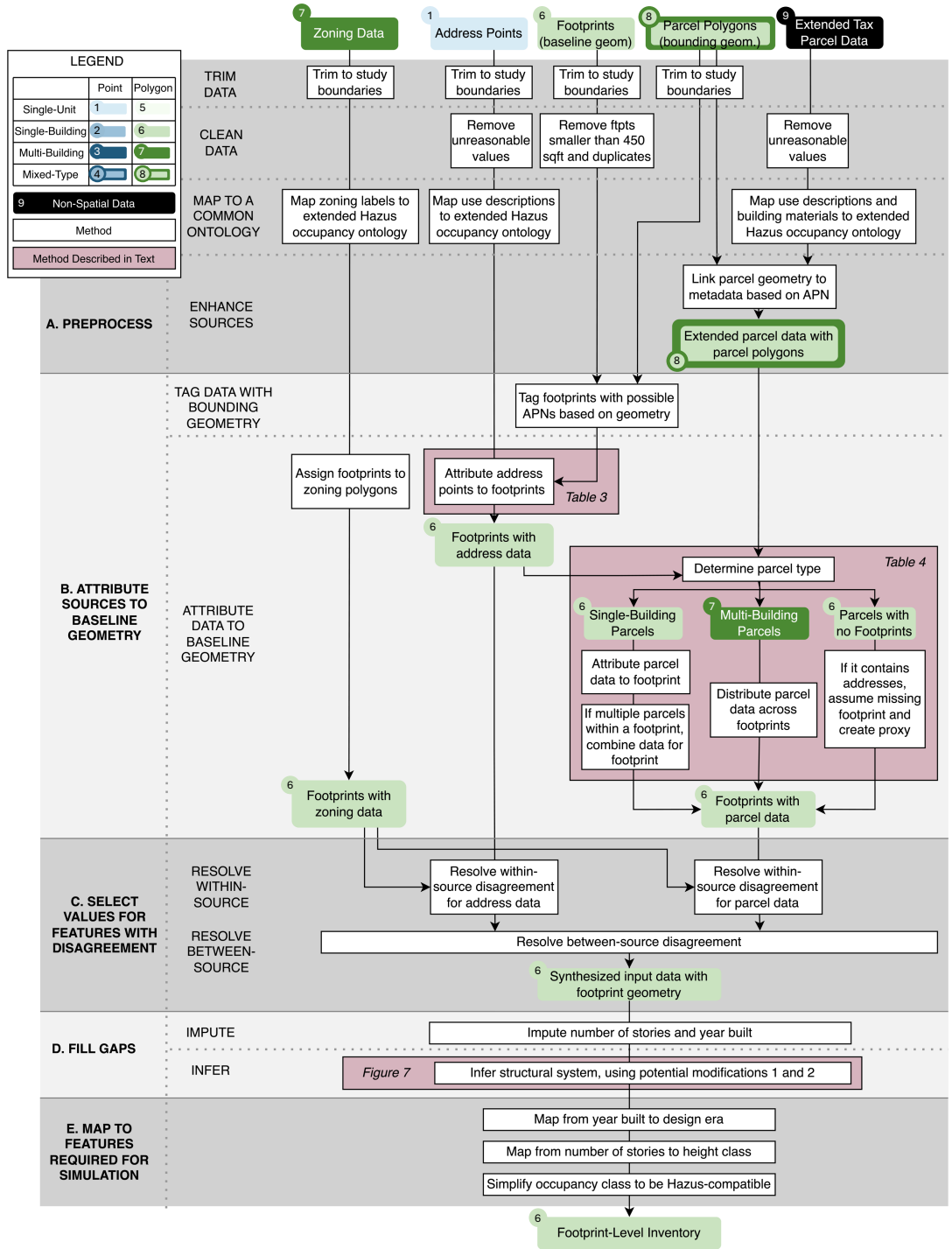
LEGEND
Point
Polygon
Single-Unit
Single-Building
Multi-Building
Mixed-Type
Non-Spatial Data
Method
Method Described in Text
7 Zoning Data
1 Address Points
6 Footprints (baseline geom)
8 Parcel Polygons (bounding geom.)
9 Extended Tax Parcel Data
TRIM DATA
Trim to study boundaries
Trim to study boundaries
Trim to study boundaries
Trim to study boundaries
CLEAN DATA
Remove unreasonable values
Remove ftpts smaller than 450 sqft and duplicates
Remove unreasonable values
MAP TO A COMMON ONTOLOGY
Map zoning labels to extended Hazus occupancy ontology
Map use descriptions to extended Hazus occupancy ontology
Map use descriptions and building materials to extended Hazus occupancy ontology
A. PREPROCESS
ENHANCE SOURCES
Link parcel geometry to metadata based on APN
Extended parcel data with parcel polygons
B. ATTRIBUTE SOURCES TO BASELINE GEOMETRY
TAG DATA WITH BOUNDING GEOMETRY
Tag footprints with possible APNs based on geometry
ATTRIBUTE DATA TO BASELINE GEOMETRY
Assign footprints to zoning polygons
Attribute address points to footprints
Table 3
Footprints with address data
Table 4
Determine parcel type
Single-Building Parcels
Multi-Building Parcels
Parcels with no Footprints
Attribute parcel data to footprint
If multiple parcels within a footprint, combine data for footprint
Distribute parcel data across footprints
If it contains addresses, assume missing footprint and create proxy
Footprints with zoning data
Footprints with parcel data
C. SELECT VALUES FOR FEATURES WITH DISAGREEMENT
RESOLVE WITHIN-SOURCE
Resolve within-source disagreement for address data
Resolve within-source disagreement for parcel data
RESOLVE BETWEEN-SOURCE
Resolve between-source disagreement
Synthesized input data with footprint geometry
D. FILL GAPS
IMPUTE
Impute number of stories and year built
INFER
Figure 7
Infer structural system, using potential modifications 1 and 2
E. MAP TO FEATURES REQUIRED FOR SIMULATION
Map from year built to design era
Map from number of stories to height class
Simplify occupancy class to be Hazus-compatible
Footprint-Level Inventory